

CASE STUDY

Designing an Automated Learning & Compliance Data Platform

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Executive Summary

Consolidating Disconnected Databases from multiple sources into
One Reliable Source of Truth

Client	Industry	Project Duration	My Role
Leading Food & Beverage Manufacturer	Manufacturing	~3 months	Operations Data & BI Consultant

Team Size	Business Area	Project Type
Cross-functional Team	Health and Safety HR Training & Compliance	Data Architecture, ETL & Business Intelligence

1050
Employees

7+
Disconnected
Databases Integrated

100%
Automated
Compliance Monitoring

Real-Time
Executive Dashboard Visibility

120
Training programs

Core Capabilities

- ETL Pipeline Design
- Operational Data Architecture
- Data Cleansing & Standardization
- Workflow Automation
- Business Intelligence & KPI Reporting
- Data Governance & Master Data Management

Context

A leading food and beverage manufacturer managed its Health & Safety (H&S) training program through multiple disconnected databases maintained by different business functions. Employee information, training requirements, attendance records, organizational structures, and compliance rules were stored independently, making it difficult to monitor mandatory training, ensure legal compliance, and provide leaders with accurate visibility into workforce readiness.

The objective of this project was to design an automated Learning & Compliance data platform capable of integrating all operational datasets into a centralized architecture. The solution would automate training compliance monitoring, reduce manual administrative work, and provide real-time visibility into organizational training status through interactive dashboards.

Centralizing and standardizing operational information into a single source of truth before reporting, significantly improves accuracy by creating more reliable KPIs, reduces manual effort, and improves decision-making.



The Challenge

The Health & Safety training process depended on several independent databases that were manually maintained and disconnected from one another.

As a consequence, compliance reporting became reactive instead of proactive, leadership lacked reliable operational visibility, and significant administrative resources were consumed maintaining disconnected information instead of supporting strategic decision-making.

Critical information such as employee records, organizational structures, mandatory training matrices, historical attendance, certification validity periods, and training schedules existed in separate files without automated relationships.

- **Training compliance had to be verified manually.**
- **Leaders had limited visibility into their teams' training status.**
- **Scheduling required significant administrative effort.**
- **Compliance indicators were difficult to calculate.**
- **Data quality issues frequently appeared due to duplicated or inconsistent information.**
- **One full-time employee spent most of their time consolidating and validating information instead of adding strategic value.**



THE APPROACH

Rather than building another reporting solution, I designed a scalable operational data architecture supporting the entire Health & Safety training lifecycle.

The approach focused on creating a centralized relational data model, implementing automated ETL processes, standardizing business rules, and establishing data governance practices that ensured long-term scalability.

This allowed operational leaders to move from manual data validation to proactive compliance management supported by real-time business intelligence.

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Solution Desing

Implemented

- Designed the complete data architecture for an automated Learning & Compliance platform.
- Built an ETL workflow integrating multiple operational databases into a centralized master dataset.
- Developed automated data transformation processes using Python (Pandas) in Google Colab.
- Standardized employee, organizational, and training information through data cleansing and normalization.
- Created relational datasets connecting employees, departments, job positions, mandatory training matrices, historical attendance, validity periods, and future training schedules.
- Automated cross-database validation using employee IDs as primary keys.
- Implemented Google Apps Script automations to synchronize information and minimize manual updates
- Built dynamic data models in Google Sheets using QUERY, IMPORTRANGE, ARRAYFORMULA, FILTER, UNIQUE, INDEX/MATCH (o XLOOKUP si aplica), Pivot Tables, Named Ranges y Conditional Formatting.
- Developed executive dashboards in Looker Studio providing real-time visibility into compliance, training coverage, overdue certifications, and workforce readiness.
- Designed a scalable architecture that automatically updates with employee onboarding, offboarding, new training programs, and attendance registrations.

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Skills Demonstrated

Data Governance

- Master Data Management
- Data Quality Management
- Data Cleansing
- Data Standardization
- Data Validation
- Business Rules Design

Business Analysis

- Business Process Analysis
- Requirements Gathering
- Operational Process Mapping

Project Management

- Stakeholder Management
- Agile Project Coordination
- Cross-functional Collaboration

Data Analytics

- Business Intelligence
- KPI Design
- Executive Reporting
- Data Visualization

Automation

- Workflow Automation
- Process Automation
- Reporting Automation

Data Engineering

- ETL Design
- Data Integration
- Data Architecture
- Data Modeling
- Relational Data Design



Tools Used

Data Processing

- Python
- Google Colab
- Pandas
- NumPy

Data Management

- Google Sheets
 - QUERY
 - ARRAYFORMULA
 - IMPORTRANGE
 - FILTER
 - UNIQUE
 - Pivot Tables
 - Conditional Formatting
 - Data Validation
 - Named Ranges

Automation

- Google Apps Script

Project Management

- ClickUp

Collaboration

- Google Workspace

Business Intelligence

- Looker Studio

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DATA ARCHITECTURE

Designing an Automated Learning & Compliance Data Platform



THE RESULTS

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The project transformed a fragmented training administration process into a fully integrated Learning & Compliance Intelligence Platform.

By centralizing operational data, automating ETL workflows, and implementing standardized business rules, the organization achieved real-time visibility into mandatory training compliance while significantly reducing administrative effort.

The scalable architecture now supports continuous workforce changes and enables operational leaders to proactively manage training performance through live business intelligence dashboards.

Business Impact

- Reduced manual administrative effort dedicated to training management by 60%, eliminating nearly 1 FTE.
- Automated ETL workflows for employee records, training matrices, attendance history, and certification validity.
- Enabled real-time monitoring of mandatory H&S training compliance.
- Improved legal compliance through automated tracking of certification validity and required training.
- Increased data quality by implementing standardized validation, cleansing, and business rules.
- Created a scalable architecture that automatically supports employee onboarding, offboarding, new training programs, and attendance updates.
- Empowered managers with self-service dashboards to identify training gaps and schedule workforce training proactively.

If your business is managing information across multiple spreadsheets or disconnected sources, - **let's build a single source of truth that supports faster and more confident operational decisions.**

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